

Justin Burzachiello

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Education

University of California, San Diego (UC San Diego) **01/2025–Present**
Ph.D. in Mechanical and Aerospace Engineering (3.49 GPA)

Stony Brook University (SBU) **08/2023–12/2024**
M.S. in Computational Applied Mathematics (3.63 GPA)

University of California, Riverside (UC Riverside) **09/2019–06/2023**
B.S. in Physics (3.80 GPA, Magna Cum Laude): Applied Physics concentrated into Computer Science

Coursework

Math : Numerical analysis/optimization, ODEs/PDEs (numerical, analytical),
Complex analysis, Measure theory, Functional analysis, Discrete math

CS : Parallel computing, Machine learning, Reinforcement learning, Computer
vision, DSA, Software engineering

Engineering : CFD, Design optimization, Model reduction, Robotics, Neuroimaging

Physics : Plasma waves, Magnetic confinement fusion, Mathematical methods,
Density functional theory

Skills

Programming : Python (PyTorch, CuPy, OpenCV, scikit-image, scikit-learn, NetworkX),
C++, MATLAB, LabVIEW

Engineering : VGSTUDIO MAX, ImageJ, EGSnrc, SOLIDWORKS

Other : MPI, LaTeX, UNIX, Bash, Slurm, Git, GitHub

Publications

Spacecraft heat shield study in the DIII-D tokamak (arXiv, **2026**)

Structural and Dynamical Analyses of Apo and Cap-binding eIF4E: An in silico Study (bioRxiv, **2024**)

Experience

UC San Diego: Orlov Lab Research Group **12/2025–Present**
Graduate research assistant (advised by Dr. Dmitri Orlov) *La Jolla, CA*

- Developing stereoscopic imaging and computer vision tools for surveillance of UFOs in DIII-D.
- Using Reeb graphs to analyze fundamental periodic orbits of chaotic PDEs.
(poster accepted at 2026 Sherwood Fusion Theory Conference).

UC San Diego: Kramer Research Group **01/2025–12/2025**
Graduate research assistant (advised by Dr. Boris Kramer) *La Jolla, CA*

- Implemented spectral submanifolds for nonlinear dimensionality reduction in Lagrangian
structure-preserving Operator Inference-based model reduction.
- Used computational graph transformations to significantly reduce computational complexity of
multidimensional numerical integration for polynomial chaos expansion uncertainty quantification.

VJ Technologies	05/2024–08/2024
<i>CT System Engineer Intern</i>	<i>Bohemia, NY</i>
- Modeled chemical filtration through FEM simulation of water flow through a CT scan of coffee. - Quantified manufacturing defects in 3D prints with CT & VGSTUDIO MAX. - Computed ionizing photonic & electronic radiation dose from particle accelerators with EGSnrc. - Used ImageJ to compute root growth increase after adding EPM to <i>Arabidopsis thaliana</i>.	
SBU: Deng Research Group	02/2024–12/2024
<i>Graduate research assistant</i> (advised by Dr. Yuefan Deng)	<i>Stony Brook, NY</i>
- Lead my team in processing CT scans of plant roots; segmented voxels-of-interest. - Analyzed molecular dynamics distance matrix time series of eIF4E protein for AI drug discovery. - TA for AMS 210–Applied Linear Algebra, AMS 361–Applied Calculus IV: Differential Equations.	
UC Riverside: Coh Research Group	03/2022–12/2023
<i>Research Assistant I in Mech. Eng.</i> (under Dr. Sinisa Coh)	<i>Riverside, CA</i>
- Created Python library for WYSIWYG tensor arithmetic from strings for quantum physics. - Created private Sphinx-based database website for hosting Wannier90/Wannier-Berri DFT data. - Used supercomputer Quantum ESPRESSO DFT software to calculate properties of materials.	
U.S. DOE, ORNL: Application Engineering Group	06/2022–08/2022
<i>Software engineer intern</i> (under Drs. Greg Watson and Ben O'Neill)	<i>Oak Ridge, TN</i>
- Documented source code of VnV verification and validation software. - VS Code extension: https://marketplace.visualstudio.com/items?itemName=jburz2001.vnv-snippets - Visualized data for finite element method mesh refinement studies.	
UC San Diego: Kramer Research Group	06/2021–09/2021
<i>Undergraduate research assistant</i> (under Dr. Boris Kramer)	<i>Remote</i>
- Configured genetic programming AI for symbolic regression-based physics derivation (1 of 13 honorees, out of 400+, to present at UC San Diego 2021 Summer Research Conference). - Documented 5 Lagrangian mechanical systems' equations, figures, and physics with LaTeX. - Used MATLAB to program ODE simulations for a mass-spring-damper and discretized linear beam.	
UC Riverside: Microwave Nano-Electronics Laboratory	02/2020–11/2021
<i>Undergraduate research assistant</i> (under Dr. Yongtao Cui)	<i>Riverside, CA</i>
- Created an FPGA-based PLL with a lock-in amplifier, PI controller, and I/O with LabVIEW (1 of 5 honorees, out of 20+; presented at UC LEADS 2021 Research Conference). - Built custom atomic force microscope (AFM) components with SOLIDWORKS. - Analyzed microscale topography with a Bruker AFM; post-processed the data with NanoScope.	

Presentations

68th Annual Meeting of the APS Division of Plasma Physics	11/2026
(Acceptance pending) Presenting poster on cross-spectral stereoscopic imaging of particulate transport in DIII-D.	
Cal-Bridge Fall Conference	09/2026
Giving a talk on stereoscopic imaging of particulate transport in DIII-D.	
MagNetUS Workshop	08/2026
Presenting poster on stereoscopic imaging of particulate transport in DIII-D.	
Sherwood Fusion Theory Conference	04/2026
Colleague presented our poster on Reeb graph analysis of plasma turbulence.	

SBU Garcia Scholars	06/2024
• Presented poster on using micro-CT with image processing to quantify plant root growth.	
Oak Ridge National Laboratory (ORNL) Summer Intern Symposium	08/2022
• Presented VnV software-based web app for ASGarD software-based numerical fluid simulation	
UC San Diego STARS Closing Ceremony	08/2021
• Nominated to present physics-based artificial intelligence research at STARS closing ceremony 1 of 2 nominees out of 60+ students	
UC San Diego 2021 Summer Research Conference	08/2021
• Conveyed how to use artificial intelligence to derive physical laws from experimental data 1 of 13 “Student Spotlight” honorees out of 400+ students	
MSRIP/UC LEADS/Cal Pre-Doc Symposium	08/2021
• Conveyed how to use artificial intelligence to derive physical laws from experimental data	
UC LEADS 2021 Koret Leadership and Research Symposium	03/2021
• Presented FPGA-based quartz tuning fork sensor for atomic force microscopy 1 of 3 “Honorable Mentions” in engineering out of 20+ students	
UC LEADS 2020 Summer Symposium	07/2020
• Presented methodologies for developing an LIA and optimizing FPGA memory consumption	
Geospatial data science: various venues	08/2017–08/2019
• Advocated for combatting homelessness with a custom ArcGIS-based resource guide: * Redlands City Hall : conveyed the need for more haircut/shower resources for the homeless * San Diego Convention Center : demoed resource guide at 2018 ESRI User Conference * University of Redlands : answered questions from San Bernardino County foster youth * Redlands Country Club : demoed resource guide to the Women’s Council of Realtors * Redlands Mitten Building : demoed resource guide to the YouthHope Foundation	

Outreach & Service

<i>Advances in Computational Mathematics</i>	2026
<i>Manuscript reviewer</i>	<i>UC San Diego</i>
• Reviewed 1 article submission for 2 rounds of reviews	
<i>UC Riverside Undergraduate Research Journal</i>	09/2021–06/2023
<i>Editorial board member</i>	<i>UC Riverside</i>
• Reviewed 6 article submissions for 3 rounds of reviews	
Mentor Collective	08/2021–06/2023
<i>Mentor</i>	<i>UC Riverside</i>
• Mentored 8 undergraduate freshmen in their first year of college	

Honors

UC Riverside	
Outstanding 1st-, 2nd-, and 4th-Year Undergraduate Student	06/2020, 2021, 2023
Graduate Deans’ Leadership Award	03/2022
Schuler Endowed Scholarship for Guardian Scholars (i.e., for foster youth)	2022–2023 academic year
UC Riverside Honors (offered)	2020
UC San Diego	
UC LEADS Fellow	2025
Cal-Bridge Doctoral Scholar	2025
Graduate Research Mentorship Program (department nominee)	2026

Attended Summer Schools, Workshops, & Conferences

68th Annual Meeting of the APS Division of Plasma Physics • 5-day conference on plasma physics.	11/2026
2026 Cal-Bridge Fall Conference • 3-day conference on professional development for promising, disadvantaged Ph.D. students.	09/2026
2026 MagNetUS Workshop • 4-day conference on plasma physics for magnetic confinement fusion.	08/2026
2026 DIII-D Summer School • 3-day summer school on engineering relevant to fusion at DIII-D.	07/2026
2026 MHD Workshop • 3-day conference on magnetohydrodynamics.	06/2026
2026 Joint IAEA/ENEA/EMFCSC Fusion School • 5-day summer school on magnetic confinement fusion engineering.	06/2026
2025 NextProf Pathfinder Workshop • 3-day workshop on professional development for promising future professors.	10/2025
2025 Structure-Preserving Scientific Computing and ML Summer School • 8-day summer school on physics-informed machine learning.	06/2025
2024 Model Reduction and Surrogate Modeling (MORe2024) • 5-day conference on model order reduction for computational physics.	09/2024